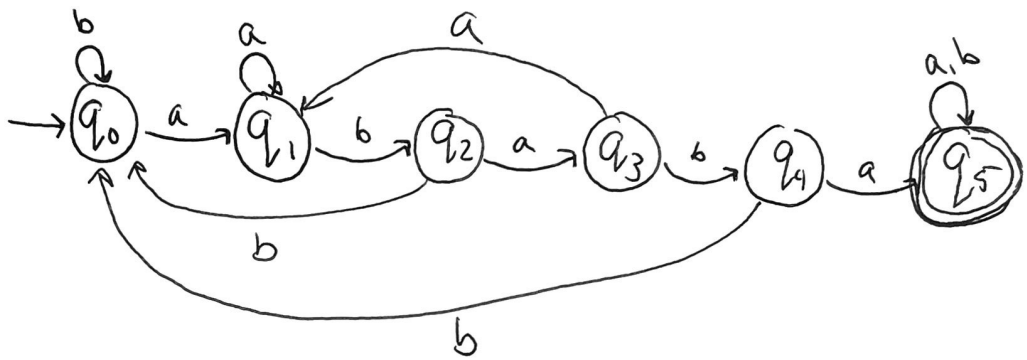
Zac Cross
CS520
10-9-2025

Homework 1

Let $\Sigma = \{a, b\}$

1.) $\{w \mid w \text{ contains the substring } ababa\}$

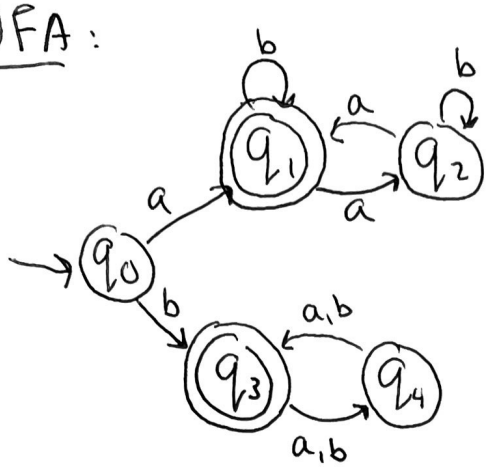
DFA:



Regular Expression: $\Sigma^* ababa \Sigma^*$

2.) $\{w \mid w$ starts with a and has an odd number of a's, or starts with b and has odd length$\}$

DFA:



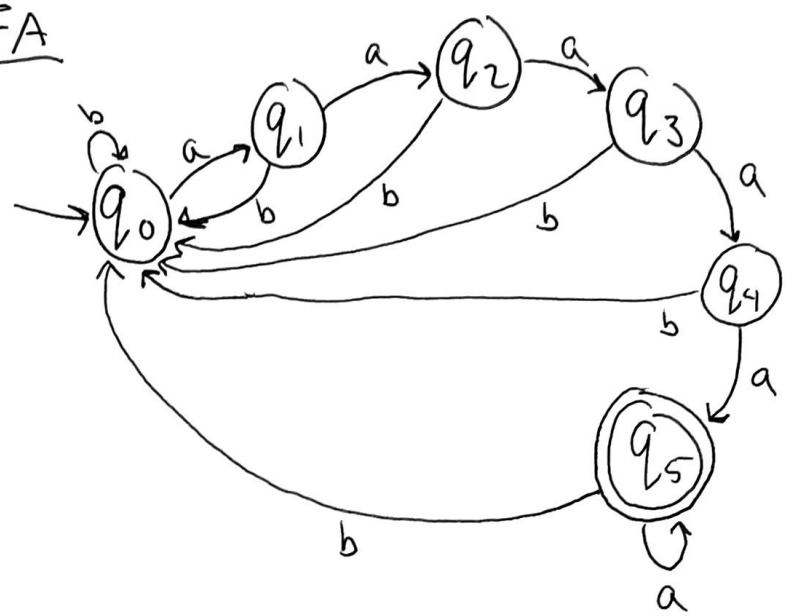
Regular Expression:

$$a(b^*ab^*a)^* \cup b(\Sigma\Sigma)^*$$

3.)

$\{w \mid w \text{ ends with } a^5\}$

DFA

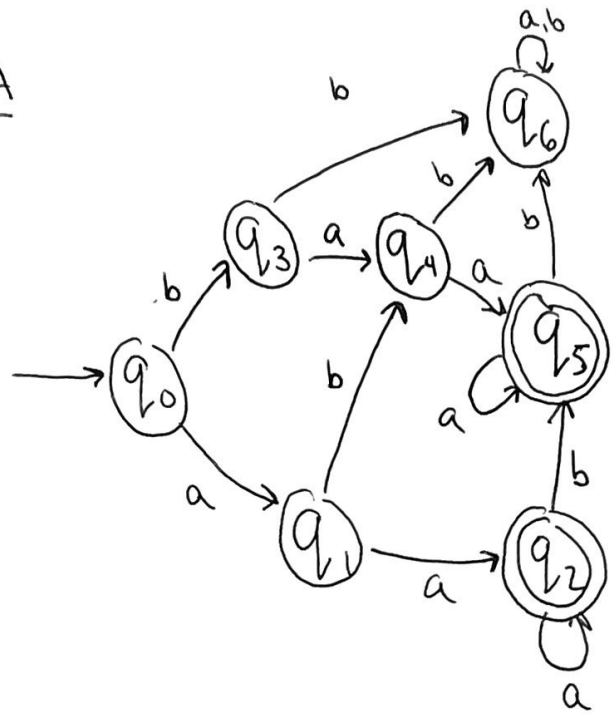


Regular Expression

$$\Sigma^* a^5 = \Sigma^* aaaaa$$

4.) $\{w \mid w$ contains at <u>least</u> 2 a's and at <u>most</u> 1 b$\}$

<u>DFA</u>



<u>Regular Expression</u>

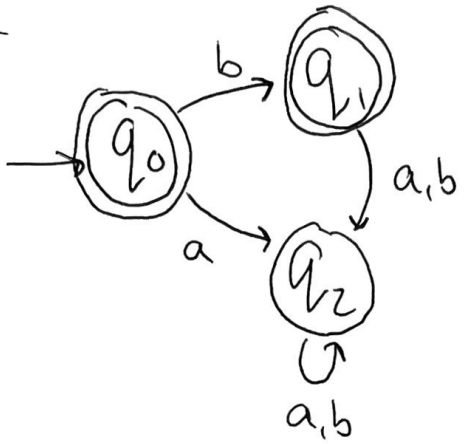
$$(\Sigma^* a)\Sigma^* a \Sigma^* \cap (a^* b a^* \cup a^*)$$

$$\equiv (\Sigma^* a \Sigma^* a \Sigma^*) \cap (a^* b^{0,1} a^*)$$

where $b^{0,1}$ is either 0 b's or 1 b.

5.) $\{\epsilon, b\}$

DFA



Regular Expression

$b \cup \varepsilon$